# Ch 12.1, 12.4: Unsupervised Learning & Clustering

## Lecture 31 - CMSE 381

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::
Dept of Computational Mathematics, Science & Engineering

Mon, Dec 5, 2022

# Announcements

**Last time:**

- Convolutional Neural Nets

**This lecture:**

- Clustering (Just hierarchical clustering)

| | | | | | |
|---|---|---|---|---|---|
| 20 | F | Nov 4 | Polynomial & Step Functions. | 7.1,7.2 | |
| 21 | M | Nov 7 | Step Functions | 7.2 | |
| 22 | W | Nov 9 | Basis functions, Regression Splines | 7.3,7.4 | |
| 23 | F | Nov 11 | Decision Trees | 8.1 | HW #7 Due |
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| 30 | W | Nov 30 | Multi Layer NN | 10.2 | |
| 31 | F | Dec 2 | CNN | 10.3 | |
| 32 | M | Dec 5 | Unsupervised Learning & Clustering | 12.1, 12.4 | HW #10 Due |
| | W | Dec 7 | Review | | |
| | F | Dec 9 | Midterm #3 | Bring your cheat sheet and a non-internet-connected calculator | |

**Announcements:**

- HW #10 Due today
- Weds: Review - Bring questions!
- Friday: Exam
  - Content since 2nd Exam (Ch 7 and on)
  - One page (8.5x11) handwritten cheat sheet

Section 1

# Unsupervised learning

# Supervised vs Unsupervised Learning

**Supervised**

Have input data $\to X$

Have output labels $\to y$

Find some function $f(X) = \hat{y}$

Evaluate the model by
seeing how close $y$ is $\hat{y}$
how often incorrect.

**Unupervised**

No output labels

More subjective

No single goal

Often used in "exploratory data analysis"

No universally accepted measure of assessment

# Some examples of unsupervised problems

- Assay gene expression levels in 100 patients with breast cancer, looking for subgroups with similar qualities
- Online shopping: find groups of shoppers with similar browsing and purchase histories and show relevant related products.
- Search engine picking results to show

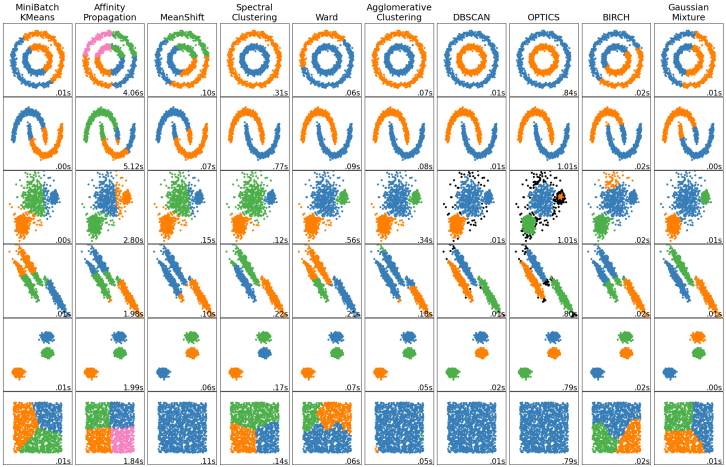
Section 2

# Clustering

# Big idea

- Find subgroups / labels in a data set
- Might specify # of clusters you want in advance

Don't care which color, just that clusters have the same color

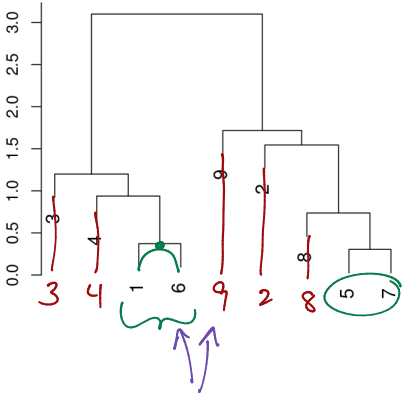


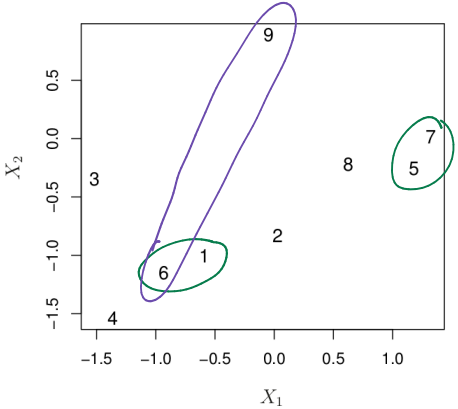
Section 3

12.4.1

# Hierarchical Clustering

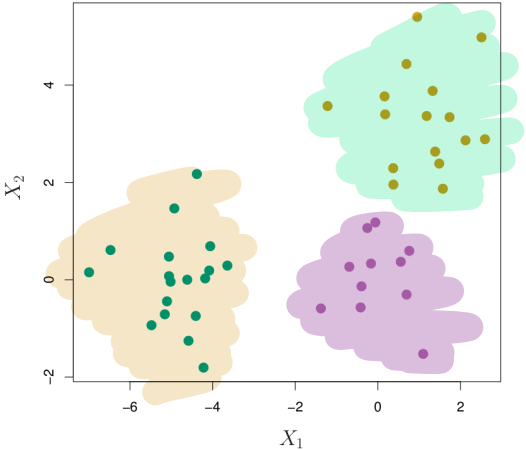
# Dendrogram

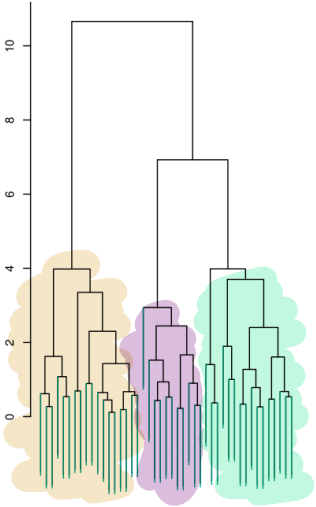


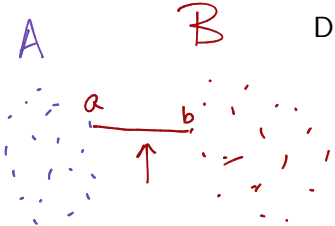

- Ordering on x-axis doesn't matter
- Similarity is based on height in y

# A bigger example



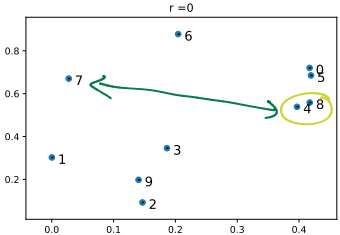



# Single linkage



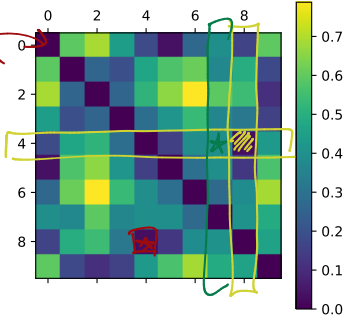

Distance between cluster $A$ and cluster $B$:
Smallest distance between the points

$$L(A,B) = \min_{a \in A, b \in B} \|a - b\|$$

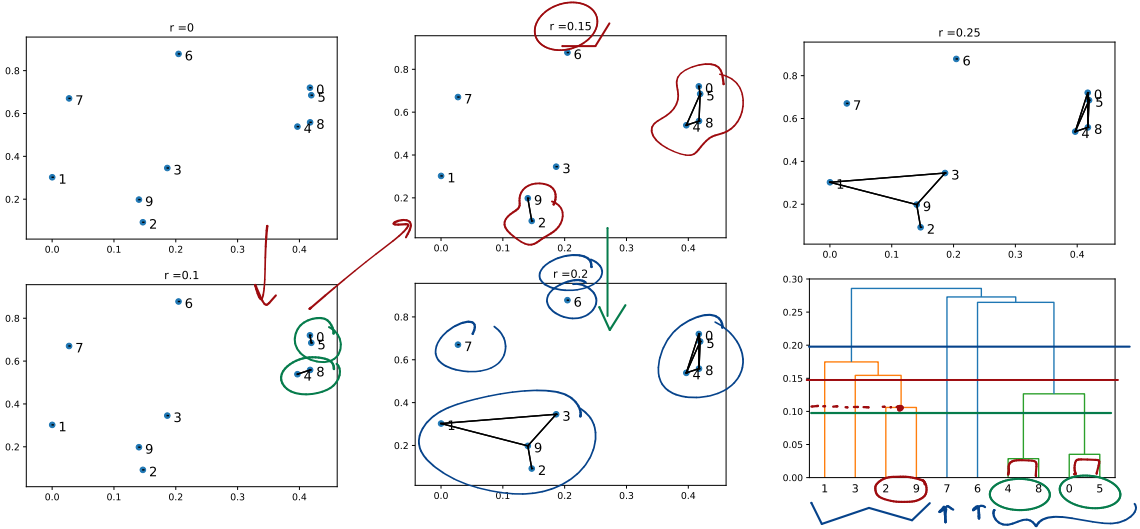



Distance matrix

Diagonal is 0

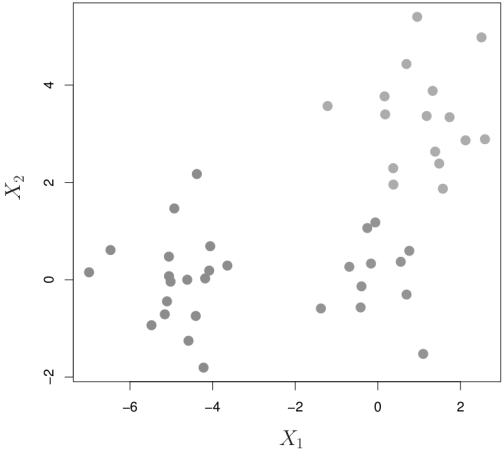


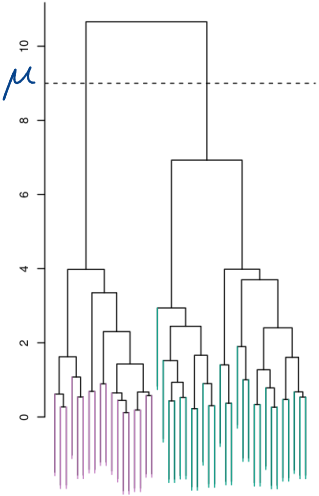
# Building the dendrogram



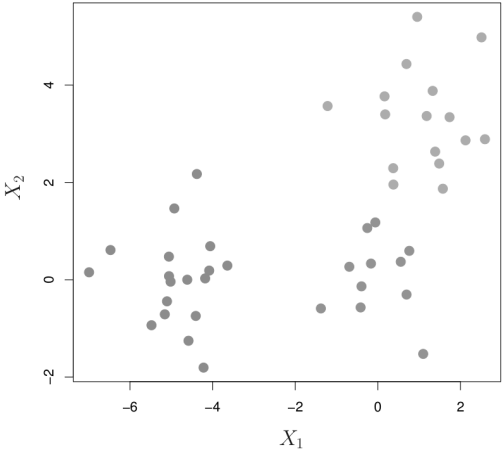
# How to get clusters

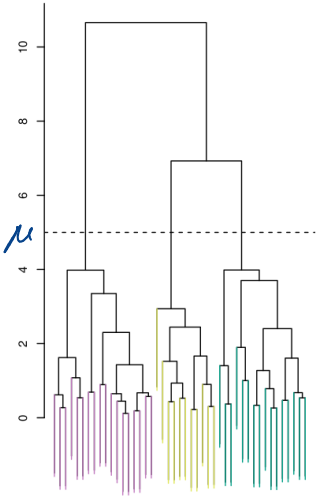
Any choice of threshold ⇝ clusters



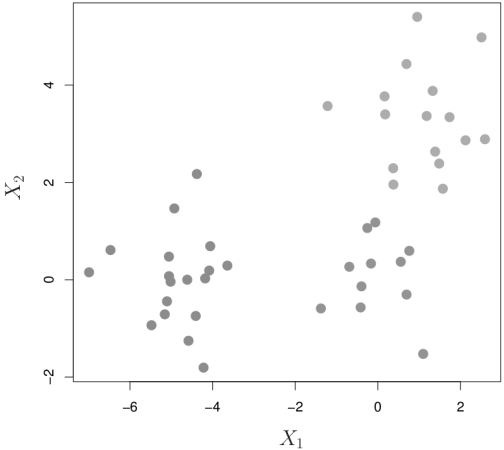



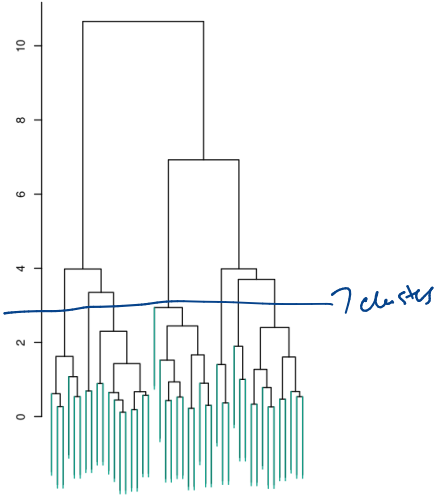
# How to get different clusters



# Can get any number of clusters



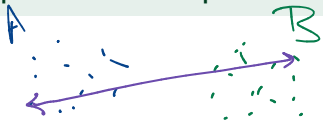



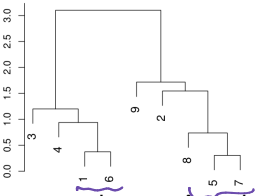
# Linkage

| *Linkage* | *Description* |
|---|---|
| Complete | Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *largest* of these dissimilarities. |
| Single | Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *smallest* of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time. |
| Average | Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *average* of these dissimilarities. |
| Centroid | Dissimilarity between the centroid for cluster A (a mean vector of length $p$) and the centroid for cluster B. Centroid linkage can result in undesirable *inversions*. |

Picture earlier

# Example with complete linkage

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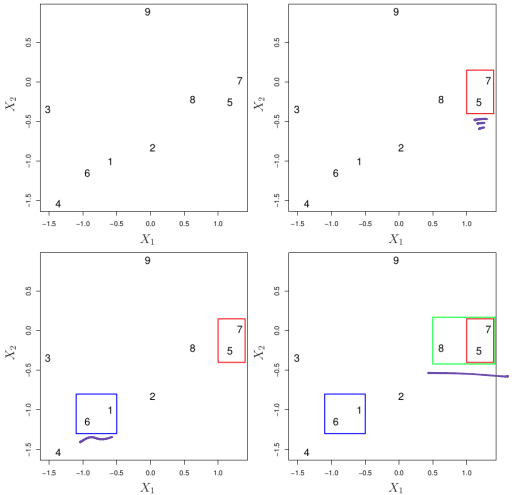
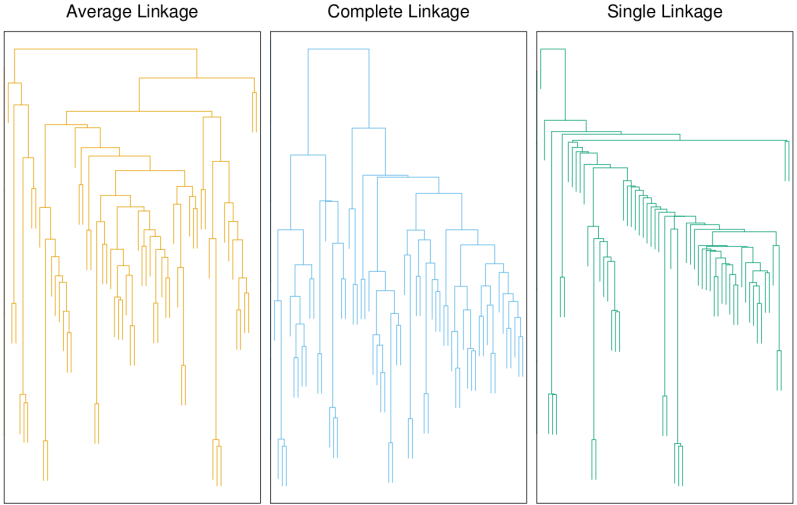


# Examples of different linkage



# Dependence on dissimilarity measure

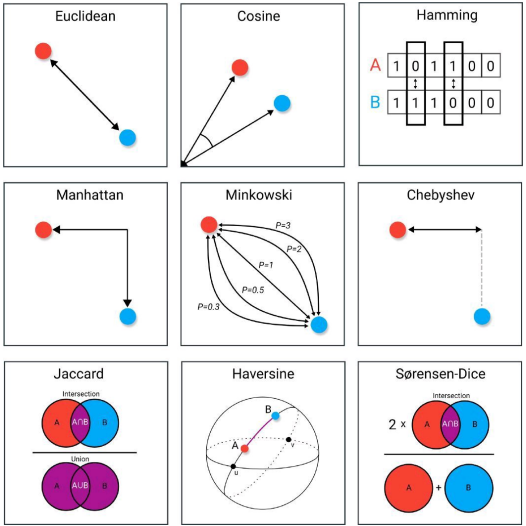



Photo Credit Link

# Coding

# Next time

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